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# Semiexperimental equilibrium molecular structure of phthalic anhydride

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ABSTRACT

The semiexperimental equilibrium molecular structure ($r_e^{se}$) of the title compound is derived from the experimental rotational constants of the parent molecule and a number of isotopologues using a quantum-chemical force field. The quadratic and cubic force fields are calculated at the B2PLYP double hybrid functional with the correlation-consistent triple-ζ cc-pVTZ basis set. The rovibrational and electronic corrections necessary for transformation of ground-state rotational constants to equilibrium ones are calculated using anharmonic (cubic) force constants and rotational $\boldsymbol{g}$ tensor, respectively. The $r_e^{se}$ structure is used to benchmark the results of coupled-cluster computations (CCSD(T)).

## 1. Introduction

The anhydride functional group consisting of two carbonyl groups connected by a single oxygen atom is rather reactive, so anhydrides are often used as chemical intermediates or for the synthesis of polymers. There are several cyclic symmetric anhydrides, for instance: maleic anhydride, $c\text{-}C_2H_2(CO)_2O$, succinic anhydride, $c\text{-}(CH_2CO)_2O$, and phthalic anhydride, $c\text{-}C_6H_4(CO)_2O$. Succinic anhydride is a saturated compound, whereas phthalic anhydride is unsaturated because an aromatic phenyl ring is attached to the anhydride group. These structural differences significantly affect the geometry of the molecule. As the equilibrium molecular structures of maleic anhydride [1,2] and succinic anhydride [3,4] are already accurately known, it is interesting to also determine an accurate equilibrium structure for phthalic anhydride. Due to the symmetry of this molecule and the subsequently small number of independent geometrical parameters, it is possible to perform a structure optimization at a high level of theory and to compute an accurate anharmonic force field used for the determination of a semiexperimental equilibrium structure. It is indeed useful to determine an equilibrium structure by two independent methods because it allows us to benchmark both methods and to have a more reliable estimate of accuracy of this structure [5].

The structure of phthalic anhydride has been previously determined by single-crystal X-ray diffraction [6]. Its vibrational spectrum has been recorded in solution and solid phases [7,8]. Recently, its microwave rotational spectrum has been measured in the centimeterwave range using a pulsed-beam Fourier transform spectrometer [9]. Rotational constants were obtained for the parent isotopologue as well as for all $^{13}$C and $^{18}$O isotopologues permitting the determination of a partial empirical structure. The molecule was assumed to be planar with $C_{2v}$ symmetry. This is confirmed by the selection rules of the observed transitions and the relatively small value of the ground state inertial defect, $\Delta_0 = -0.154$ uÅ$^2$. Ab initio structure optimizations further confirm this conclusion. The ground state inertial defect is negative indicating the existence of a low-frequency out-of-plane vibration. Using the empirical formula of Oka [10].

$$\Delta_{\text{vib}} = \frac{-33.715}{\nu_t} + 0.00803\sqrt{I_c} \tag{1}$$

One finds $\nu_t = 92$ cm$^{-1}$ not too far from the lowest-frequency value of 127 cm$^{-1}$ (MP2_FC/VTZ, see next Section for abbreviations). However, note that this empirical equation assumes that the out-of-plane vibration is isolated, which is not the case here, the next vibration being at 158 cm$^{-1}$ (MP2_FC/VTZ value).

This article is organized as follows. The next section describes the computational details. Section 3 explains how a Born-Oppenheimer (BO) equilibrium structure, $r_e^{BO}$, is computed and section 4 is devoted to the calculation of the semiexperimental equilibrium structure, $r_e^{se}$.

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**Table 1**
Ab initio and semiexperimental equilibrium structural parameters of phthalic anhydride (distances in Å, angles in degrees).

| Method<br>Basis set | MP2_FC<br>wCVTZ | MP2_AE<br>wCVTZ | MP2_FC<br>VTZ | MP2_FC<br>VQZ | CCSD(T)_FC<br>VTZ | $r_e^{BO}$ [a] | $r_e^{se}$ |
|---|---|---|---|---|---|---|---|
| C1—C2 | 1.3851 | 1.3822 | 1.3867 | 1.3849 | 1.3880 | 1.3833 | 1.3814(11) |
| C2—C3 | 1.4779 | 1.4748 | 1.4788 | 1.4774 | 1.4856 | 1.4812 | 1.4798(9) |
| C3—O4 | 1.3958 | 1.3932 | 1.3969 | 1.3946 | 1.3961 | 1.3913 | 1.3903(4) |
| C6—C2 | 1.3852 | 1.3823 | 1.3868 | 1.3847 | 1.3906 | 1.3856 | 1.3844(8) |
| C7—C6 | 1.3936 | 1.3907 | 1.3952 | 1.3930 | 1.3987 | 1.3936 | 1.3921(6) |
| C8—C7 | 1.4001 | 1.3971 | 1.4017 | 1.3993 | 1.4057 | 1.4004 | 1.3984(11) |
| C3═O10 | 1.1954 | 1.1935 | 1.1969 | 1.1942 | 1.1950 | 1.1904 | 1.1887(4) |
| C6—H12 | 1.0807 | 1.0794 | 1.0808 | 1.0802 | 1.0820 | 1.0801 | 1.0803(12) |
| C7—H13 | 1.0810 | 1.0797 | 1.0812 | 1.0805 | 1.0827 | 1.0807 | 1.0806(12) |
| C1—C2—C3 | 107.73 | 107.74 | 107.72 | 107.73 | 107.64 | 107.66 | 107.69(10) |
| O4—C3—C2 | 107.33 | 107.32 | 107.34 | 107.28 | 107.31 | 107.24 | 107.21(3) |
| C5—O4—C3 | 109.88 | 109.88 | 109.88 | 109.98 | 110.09 | 110.19 | 110.20(1) |
| C6—C2—C3 | 130.40 | 130.39 | 130.41 | 130.39 | 130.44 | 130.42 | 130.40(5) |
| C7—C6—C2 | 116.83 | 116.84 | 116.84 | 116.80 | 116.80 | 116.78 | 116.77(4) |
| C8—C7—C6 | 121.30 | 121.29 | 121.29 | 121.32 | 121.29 | 121.30 | 121.25(12) |
| C2—C3═O10 | 130.47 | 130.45 | 130.47 | 130.54 | 130.44 | 130.49 | 130.40(5) |
| C2—C6—H12 | 121.22 | 121.20 | 121.22 | 121.24 | 121.25 | 121.26 | 121.24(12) |
| C6—C7—H13 | 119.60 | 119.60 | 119.61 | 119.59 | 119.61 | 119.59 | 119.56(12) |

[a] Calculated according to Eq. (2), see text. For atom numbering see Fig. 1.

## 2. Computational details

The geometry optimizations were performed at the MP2 (second-order Møller–Plesset perturbation theory) [11] and CCSD(T) (coupled cluster theory, including single and double excitations (CCSD) [12] augmented with a perturbative estimate of the effects of connected triple excitations) [13] levels of electronic structure theory. Different atom-centered, fixed-exponent Gaussian basis sets have been utilized, including the correlation-consistent polarized basis sets cc-pVTZ and cc-pVQZ [14] abbreviated throughout this article as VTZ and VQZ, respectively. Optimizations were also performed with the correlation-consistent weighted core-valence triple-ζ, cc-pwCVTZ basis set [15] abbreviated as wCVTZ. With this last basis set, calculations were performed in the frozen core (FC) approximation as well as with all electrons correlated (AE). For calculations of the force field up to semidiagonal quartic terms, the double hybrid B2PLYP functional [16] was used together with the VTZ basis set. This level of theory was shown to give good results in anharmonic force field calculations [17]. The rotational **g**-tensor was calculated at the Hartree-Fock level of theory with the augmented correlation-consistent basis set variant including diffuse functions, aug-VTZ basis set [18]. For this calculation, the CCSD(T)_FC/TZP structure calculated in the present study was used as reference because it is close to the accurate equilibrium structure. A lower level of theory was used because a high accuracy is not needed for the **g**-tensor, its contribution being quite small.

The structure calculations were carried out with the CFOUR [19] and MOLPRO [20,21] program packages whereas the force field calculations were performed with the Gaussian16 program package [22]. The calculation of the **g**-tensor was made with CFOUR [19].

**Table 2**
Ground state rotational constants and vibrational and electronic corrections to the experimental rotational constants of phthalic anhydride (in MHz).

| Isotopic species | Axis | $B_0^{exp}$ [9] | $\Delta B_{vib}$ | $-\Delta B_{elec}$ | $\Delta B^{calc}$ [a] |
|---|---|---|---|---|---|
| Parent | *a* | 1801.7622 | 11.6864 | −0.0513 | 11.7377 |
| | *b* | 1191.7182 | 6.8145 | −0.0211 | 6.8357 |
| | *c* | 717.4461 | 4.1972 | 0.0030 | 4.1942 |
| $^{13}$C3&C5 | *a* | 1793.5560 | 11.5683 | −0.0508 | 11.6191 |
| | *b* | 1187.8460 | 6.7627 | −0.0210 | 6.7836 |
| | *c* | 714.7450 | 4.1613 | 0.0029 | 4.1584 |
| $^{13}$C1&C2 | *a* | 1798.7083 | 11.6403 | −0.0511 | 11.6914 |
| | *b* | 1191.6141 | 6.7625 | −0.0211 | 6.7836 |
| | *c* | 716.9261 | 4.1705 | 0.0029 | 4.1676 |
| $^{13}$C6&C9 | *a* | 1788.9864 | 11.5931 | −0.0506 | 11.6437 |
| | *b* | 1186.0775 | 6.7502 | −0.0210 | 6.7712 |
| | *c* | 713.3798 | 4.1603 | 0.0029 | 4.1574 |
| $^{13}$C7&C8 | *a* | 1798.7650 | 11.6487 | −0.0511 | 11.6998 |
| | *b* | 1172.9003 | 6.6944 | −0.0205 | 6.7150 |
| | *c* | 710.1185 | 4.1463 | 0.0029 | 4.1435 |
| $^{18}$O4 | *a* | 1801.8619 | 11.5695 | −0.0513 | 11.6208 |
| | *b* | 1170.3830 | 6.7362 | −0.0203 | 6.7565 |
| | *c* | 709.6783 | 4.1454 | 0.0029 | 4.1425 |
| $^{18}$O10&O11 | *a* | 1742.3734 | 11.2192 | −0.0480 | 11.2672 |
| | *b* | 1175.9439 | 6.6943 | −0.0205 | 6.7148 |
| | *c* | 702.2415 | 4.0914 | 0.0029 | 4.0885 |

[a] $\Delta B^{calc} = \Delta B_{vib} + \Delta B_{elec}$.

## 3. Ab initio Born-Oppenheimer equilibrium structure

The geometry optimizations at the CCSD(T)_FC/VTZ level of theory was performed by means of two programs (CFOUR and MOLPRO), and the results were identical. As convergence is not achieved at the CCSD(T)_FC/VTZ level, the small correction VTZ → VQZ was estimated in the MP2 approximation, see Eq. (2).

Finally, for the treatment of the inner-shell correlation effects, the MP2 method and the wCVTZ basis set were used. On the one hand, it is known that the wCVTZ basis set is too small to recover all the electronic correlation [23] but, on the other hand, the MP2 method overestimates this correction [24]. In consequence, the errors at least partially compensate each other. It is possible to check this assumption because the core correction is known to be relatively constant for a given bond. For instance, for the C—O single bond, it is about 0.0028 Å and 0.0021 Å for the C═O double bond [25] in fair agreement with the results of the present work, see Table 1. Likewise, the correction for the C—C single bond is 0.0032 Å in $C_2H_6$ [26] again in good agreement with our results, 0.0030 Å. Finally, the core correction for the C—H bond length has also the expected value, between 0.0012 Å and 0.0015 Å [23]. In conclusion, it seems that the use of the MP2/wCVTZ level of theory to estimate the core correlation does not much deteriorate the accuracy.

Summarized, the composite ab initio $r_e^{BO}$ parameters were obtained using the following equation:

$$r_e^{BO} = r_e[\mathrm{CCSD(T)\_FC/VTZ}] + r_e[\mathrm{MP2\_FC/VQZ}] - r_e[\mathrm{MP2\_FC/VTZ}] + r_e[\mathrm{MP2\_AE/wCVQZ}] - r_e[\mathrm{MP2\_FC/wCVQZ}] \quad (2)$$

The results are given in Table 1 and the corresponding Cartesian coordinates are given in Table S1 of the Supplementary Material.

## 4. Calculation of the semiexperimental structure

The semiexperimental equilibrium rotational constants were obtained from the ground state rotational constants taken from Ref. [9] after corrections for the rovibrational correction and the electronic correction [27]:

$$B_e^{\xi} = B_0^{\xi} + \Delta_{rovib}^{\xi} + \Delta B_{\mathrm{elec}}^{\xi} \quad (3)$$

where $\xi = a, b, c$ and

$$\Delta_{\mathrm{rovib}}^{\xi} = \frac{1}{2}\sum_k \alpha_k^{\xi}$$

$$\Delta_{\mathrm{rovib}}^{\xi} = -\frac{m_e}{M_p} g_{\xi\xi}, \quad where$$

$\alpha_k^{\xi}$ is the vibration-rotation interaction constant for the vibrational state $k$ and the principal axis $\xi$, $m_e$ is the mass of the electron, $M_p$ is the mass of the proton, and $g_{\xi\xi}$ is expressed in units of the nuclear magneton [28]. The rotational constants were furthermore corrected for centrifugal distortion effect. This small correction was calculated using the harmonic part of the force field.

The rovibrational corrections were calculated using anharmonic (cubic) force field (B2PLYP/VTZ) by means of the Shrink program [29,30]. The ground state rotational constants, the rovibrational corrections, and the electronic corrections are given in Table 2, whereas the semiexperimental equilibrium rotational constants are presented in Table S2 of the Supplementary Material.

It is also interesting to have a look at the inertial defect. For the parent species, the ground state value is $\Delta_0 = -0.154$ uÅ$^2$. When calculated from the semiexperimental equilibrium rotational constants, it is one order of magnitude smaller, $\Delta_e = -0.014$ uÅ$^2$. However, if the electronic correction is neglected, it is $-0.032$ uÅ$^2$, i.e. a factor two larger. It shows that the electronic correction, although small, is not negligible. The fact that the equilibrium inertial defect is not exactly zero will be the source of a small systematic error in the molecular parameters, as will be discussed below. There is a third way to check the accuracy of the semiexperimental equilibrium rotational constants: as the principal axis $a$ is a symmetry axis, the $A$ rotational constants for the parent species and the $^{18}$O4 isotopologue should be identical. Actually, it is 1813.500 MHz for the parent species and 1813.483 MHz for the $^{18}$O4 isotopologue, i.e. there is a small difference of 17 kHz.

The molecular structure is completely defined by 14 independent molecular parameters: 7 bond lengths and 7 bond angles. There are rotational constants for 7 isotopologues but, as the molecule is planar, the rotational constants are not independent and there are only 14 independent rotational constants. Furthermore, there is no isotopic substitution available for the hydrogen atoms preventing an accurate determination of their coordinates. All the moments of inertia were used in the fit. They were weighted according to their uncertainty. A first fit was made with the coordinates of the hydrogen atoms fixed at their ab initio ($r_e^{BO}$) values. At first sight, the fit is reasonably good but, actually, there are two problems: i) the system of normal equations is ill-conditioned leading to rather large standard deviations of 0.003 Å for two parameters (the C2—C3 and C2—C6 bond lengths), ii) the residuals are not random: they are all negative for the $A$ and $B$ rotational constants and all positive for the $C$ constants. There are two possible explanations for the second problem: fixing the internal coordinates of the hydrogen atoms may induce some bias or, more likely, it is due to the fact that the semiexperimental equilibrium inertial defect is not exactly zero. The first difficulty is also easy to explain: there are only 14 independent data to determine 10 parameters and the atoms C1 and C2 have very small Cartesian coordinates: $a = 0.236$ Å and $b = 0.691$ Å. In other words, the isotopologue with $^{13}$C in position C1 (or C2) does not bring enough new information.

There is an easy way to solve these problems because the $r_e^{\mathrm{BO}}$ structure is fairly accurate and can be used as predicates in the mixed estimation method with conservative uncertainties (to estimate their weight) of 0.002 Å for the bond lengths and 0.2° for the bond angles [31]. This considerably improves the conditioning but the residuals are still not random confirming that this discrepancy is due to the non-zero inertial defect. However, it is known that the rovibrational correction is affected mainly by a systematic error, which is a few percent of the rovibrational correction [32]. This is a common problem for planar molecules [33] but it is possible to correct for most of this error by multiplying all the rovibrational corrections by a constant factor $f$ of 1.11. This scaling factor multiplying the rovibrational corrections was chosen to reduce to a minimum the semiexperimental equilibrium inertial defect of the parent species: it was reduced from $-0.0136$ uÅ$^2$ to $-0.0004$ uÅ$^2$. With this choice the equilibrium inertial defect of the different isotopologues becomes nearly zero (mediane: $-0.0018$ uÅ$^2$ instead of $-0.0153$ uÅ$^2$). The subsequent fit is much better: the reduced standard deviation of the fit is reduced by a factor of almost two, the standard deviations of the parameters and the systematic deviations of the residuals are much smaller. However, the parameters of this new fit are almost identical to those of the previous fit, confirming that small systematic errors do not have much effect on the values of the structural

**Table 3**
Comparison of the geometrical parameters of maleic, succinic and phthalic anhydrides ($r_e^{se}$ values; distances in Å and angles in degrees).

| | maleic [1] | succinic [3] | phthalic (this work) |
|---|---|---|---|
| C—O | 1.3848(3) | 1.3824(5) | 1.3903(4) |
| C=O | 1.1894(2) | 1.1878(5) | 1.1887(4) |
| C—C | 1.4849(5) | 1.5116(3) | 1.4798(9) |
| ∠(C—O—C) | 108.52(3) | 111.32(3) | 110.20(1) |
| ∠(O=C—O) | 122.55(4) | 121.52(3) | 122.22(6) |
| ∠(O—C—C) | 107.78(2) | 109.77(5) | 107.21(3) |

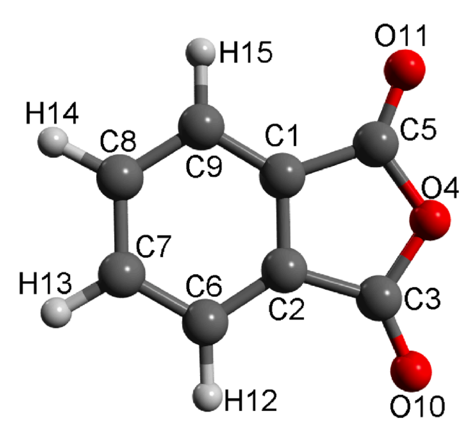


**Fig. 1.** Molecular model of phthalic anhydride with atom numbering.

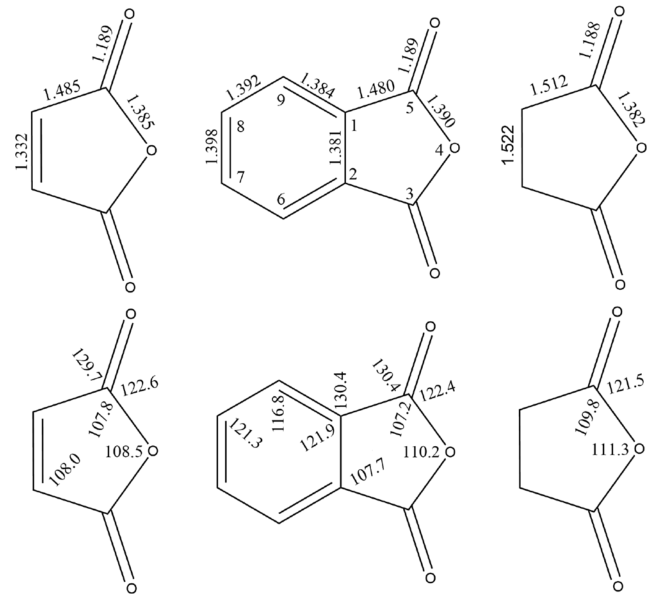


**Fig. 2.** Comparison of geometrical parameters of maleic [1], phthalic (this work) and succinic [3] anhydrides ($r_e^{se}$ values; distances in Å and angles in degrees).

**Table 4**
Atomic charges in the anhydride functional groups of maleic, succinic and phthalic anhydrides.

| | maleic | succinic | phthalic |
|---|---|---|---|
| —O— | −0.163 | −0.154 | −0.162 |
| =O | −0.266 | −0.288 | −0.276 |
| C(=O) | 0.267 | 0.269 | 0.268 |
| C(—C=O) | −0.055 | −0.140 | −0.010 |

parameters [34]. The “scaled” rotational constants and the residuals of the fit are given in Table S2 of the Supplementary Material and the derived parameters in Table 1. The semiexperimental equilibrium Cartesian coordinates are given in Table S3 of the Supplementary Material. The $r_e^{se}$ structure is in pleasing agreement with the $r_e^{BO}$ structure. However, the $r_e^{BO}$ bond lengths between heavy atoms are 0.001–0.002 Å longer than the $r_e^{se}$ ones. It may be explained by the fact that the structure is not yet fully converged at the quadruple zeta level and that the core correction is still slightly too small. From this comparison of the two structures, we may conclude that the accuracy of the $r_e^{BO}$ structure is not worse than 0.002 Å for the bond lengths and as good as 0.2° for the bond angles.

## 5. Conclusions

An accurate semiexperimental equilibrium geometry was determined for phthalic anhydride. This geometry is shown to be slightly more accurate than the $r_e^{BO}$ ab initio structure. However, the comparison of the two structures allows us to estimate the accuracy of the $r_e^{BO}$ structure, which has the expected values, i.e. 0.002 Å for the bond lengths and 0.2° for the bond angles [35–37], the $r_e^{BO}$ bond distances between heavy atoms being systematically too long. This behavior is expected for the bond lengths calculated at the CCSD(T) level with a basis set of quadruple-ζ quality (see also Ref. [5]). It is shown that the accuracy of the parameters is not very sensitive to the small systematic error affecting the rotational constants.

The equilibrium structures of the anhydride functional group in maleic anhydride, succinic anhydride, and phthalic anhydride are compared in Table 3 (see also Fig. 2). They are quite similar and to point out their small differences, accurate equilibrium structures are required. The C—O single bond in phthalic anhydride is the longest one whereas the C—C single bond, at 1.480 Å, is the shortest one, compared to maleic anhydride, 1.485 Å, and succinic anhydride, 1.512 Å.

In order to have a better understanding of the bonding, the atomic charges were calculated with the recent Charge Model 5 (CM5) of Marenich et al. [38]. The results are given in Table 4. The differences are small. However, it seems to be obvious that the C—O and C—C bond lengths decrease as the polarity of the bond increases.

*CRediT authorship contribution statement*

**Alexander V. Belyakov:** Conceptualization, Investigation, Data curation, Writing – review & editing, Visualization. **Natalja Vogt:** Conceptualization, Investigation, Data curation, Writing – review & editing, Visualization. **Jean Demaison:** Conceptualization, Investigation, Writing – original draft. **Roman Yu. Kulishenko:** Formal analysis, Software. **Alexander A. Oskorbin:** Formal analysis, Software.

## Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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## Appendix A. Supplementary material

Supplementary data to this article can be found online at https://doi.org/10.1016/j.cplett.2022.139540.

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